## Bohmian Pilot waves

$$i\hbar\frac{d}{dt} = -\frac{\hbar}{2m}\nabla^2\psi + U\psi$$

↳ * $\psi = \underbrace{Re^{iS/\hbar}}_{\varnothing}$

• in an eq$^{\underline{n}}$ w/ complex numbers can split into 2

$$\begin{cases} -R\frac{\partial S}{\partial t} = -\frac{\hbar}{2m}\nabla^2 R + \frac{1}{2m}R|\nabla S|^2 + UR & \text{similar to velocity} \\ \frac{\partial R}{\partial t} = -\frac{1}{m}(\nabla R)\cdot(\nabla S) - \frac{1}{2m}R\nabla^2 S & \text{similar to phase} \end{cases}$$

$\rho = R^2$ *probability 'density'

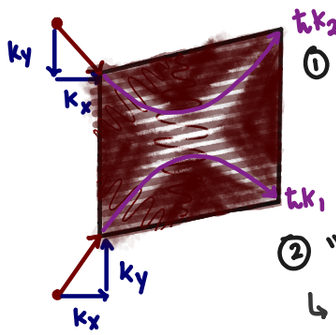
$\vec{v} = \frac{\nabla S}{m}$

$$\frac{\partial S}{\partial t} = \underbrace{-\frac{\hbar^2}{2m}\frac{\nabla^2 R}{R}}_{Q} + \underbrace{\frac{1}{2m}|\nabla S|^2}_{KE} + \overbrace{U}^{PE}$$

looks like Schrödinger

Bohmian Term
"potential of something"

interpretation: both physical wave & particles




problems:

① 2 types of particles
$\hbar k$ - 'quanta'
and
"Bohmian particles"

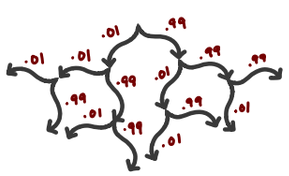
② "weird behavior" of Bohmian particles (don't appear to conserve momentum)
↳ can't cross nodes

③ only accurate for particles with mass, not accurate for photons, phonon

④ only solved for non-relativistic equation

⑤ doesn't say where randomness comes from

## Back 2 Many-Worlds

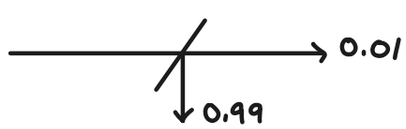



- what are the equations / Postulates of QM

• matter waves
↓
Schrö
Dirac

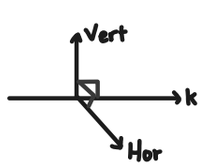
Born Rule for probabity

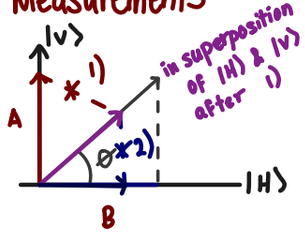
$P(\text{measurement}) \rightarrow |\psi|^2$




## Successive Measurements

polarization




• 50% gets through after measurement 1
↳ ~0% after 2nd

1) start with |V⟩ - sets all vert.

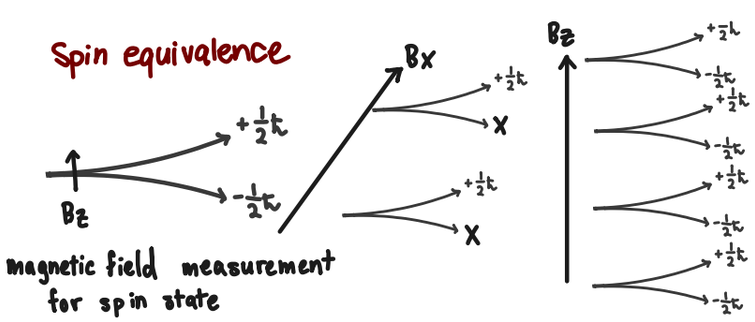
2) polarizer @ 45°
→ kill $E_\perp$
keep $E_{\parallel}$ → @ 45°

3) 2nd polarizer along |H⟩

→ remaining part is along |H⟩

* inserting 45° polarizer in between 1) and 2) results in greater flux at end

## Spin equivalence

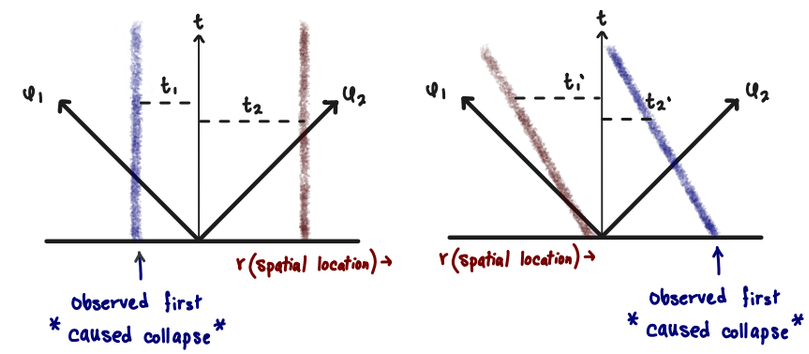


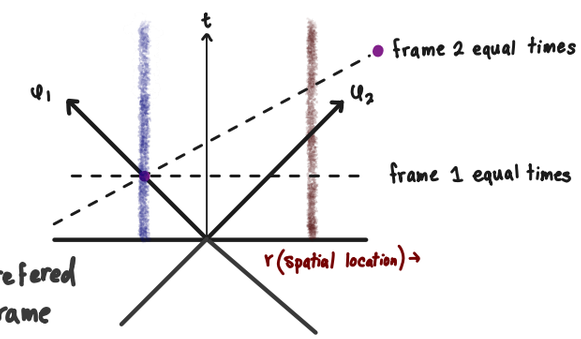

magnetic field measurement for spin state

measurement leads to projection of states onto the set of possible measurement outcomes

## "Spontaneous Collapse"

entangled ℓ relativistic ψ collapse explanation




postulate: universal prefered reference frame

× copernican principle, no 'special' reference frame

→ not a 'narrativity', no unique predictions

→ future events can affect past in other reference frame